Practical Study of Machine Learning Classifiers on Iris Species Data

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1. Introduction

Machine Learning classification is a supervised learning approach used to categorize data into predefined classes. This project focuses on implementing and comparing three classification algorithms: Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and an additional algorithm (Decision Tree).

The Iris dataset is used, which contains measurements of flower features such as sepal length, sepal width, petal length, and petal width. The goal is to predict the species of the flower.

2. Dataset Description

The Iris dataset consists of:

150 samples

3 classes (Setasoa, Versicolor, Virginica)

4 features:

Sepal Length

Sepal Width

Petal Length

Petal Width

This dataset is widely used for classification tasks due to its simplicity and effectiveness.

3. Algorithms Used

3.1 Support Vector Machine (SVM)

SVM is a supervised learning algorithm used for classification tasks. It works by finding the optimal hyperplane that separates data points of different classes.

Kernels Used:

- Linear Kernel
- RBF (Radial Basis Function) Kernel
- Sigmoid Kernel
- Polynomial Kernel

Advantages:

- Effective in high-dimensional spaces
- Works well with clear margin separation

Limitations:

- Not suitable for very large datasets
- Choosing the right kernel is important

3.2 K-Nearest Neighbors (KNN)

KNN is a simple, non-parametric algorithm that classifies a data point based on the majority class among its nearest neighbours.

Working Principle:

- Choose value of K
- Calculate distance from new point to all points
- Select K nearest neighbors
- Assign majority class
- Advantages:
- Simple and easy to understand
- No training phase

Limitations:

- Slow for large datasets
- Sensitive to value of K

3.3 Decision Tree (Additional Algorithm)

Decision Tree is a supervised learning algorithm used for classification and regression. It splits the dataset into branches based on feature values.

Working:

- Select best feature based on criteria (Gini/Entropy)
- Split dataset into subsets
- Repeat until leaf nodes are formed

4. Conceptual Understanding of Decision Tree

4.1 Key Assumptions

- Data can be split based on feature values
- Features are independent
- Patterns can be represented as tree structure

4.2 Strengths

- Easy to understand and interpret
- Works well with both numerical and categorical data
- No need for feature scaling

4.3 Limitations

- Can overfit the data
- Sensitive to small data changes

4.4 Suitable Scenarios

- When interpretability is important
- Small to medium datasets

4.5 Not Suitable When

- Dataset is very large
- High accuracy is required without tuning

5. Model Evaluation

All models were evaluated using:

- Accuracy Score
- Confusion Matrix

Observations:

- SVM performed well with RBF kernel
- KNN performance depended on value of K
- Decision Tree gave fast and interpretable results

6. Model Comparison

- Algorithm
- Accuracy
- Performance

SVM (RBF)

- High
- Best performance

KNN

- Depends on K
- Moderate

Decision Tree

- Good
- Easy to interpret

Analysis:

- SVM provided the highest accuracy among all models

- KNN required careful tuning of K value
- Decision Tree was easiest to understand but slightly less accurate

7. Key Learnings

- Different algorithms perform differently on the same dataset
- Kernel selection is important in SVM
- Choosing correct K is critical in KNN
- Decision Trees are powerful but can overfit
- Model comparison helps in selecting the best approach

8. Conclusion

In this study, multiple classification algorithms were implemented and compared using the Iris dataset. SVM with RBF kernel achieved the best performance, while KNN and Decision Tree also produced satisfactory results. Each algorithm has its own strengths and limitations, and the choice depends on the problem and dataset.